

Education

BTech - Robotics and Autonomous Systems	Plaksha University, Mohali	8.98 / 10	2027
12th - CBSE	Global Indian Intl. School, Abu Dhabi	95%	2023
10th - CBSE	Global Indian Intl. School, Abu Dhabi	92.4%	2021

Experience

Research Intern, UR Rao Satellite Centre, Indian Space Research Organisation (ISRO) Jun 2026 - Jul 2026

- Worked on learning & control of **legged robots for lunar exploration** under the Mission Simulation Group at ISRO.
- Built a **robot and simulator agnostic training framework** to design hierarchical locomotion controllers for sim-to-sim/real learning.
- Designed **hierarchical RL controllers** combining a learned high-level foot placement policy with optimisation-based **MPC and QP** for joint control.
- Benchmarked learnt latent **reduced-order models** against first-principles models (LIP, ALIP, SRB) through max speed and disturbance metrics.
- Trained foot-placement policies tracked via a task-space QP, achieving **<0.05 m/s velocity tracking error** and surviving **1.3m/s lateral impulses**.

Student Researcher, Dixon IoT Lab, Plaksha University Aug 2025 - Jun 2026

- **Air Quality Analysis of North India:**
 - Scraped over **30 million datapoints of live air quality data** from across north India, at a frequency of 10 minutes for over 4 months.
 - Developed a novel encoder-decoder model to **analyze spatio-temporal air quality data** and identify regional pollution trends over time.
 - Conducted a **divergence analysis** of pollution motion across North India by integrating wind data to find pollution sources and sinks.
- **Air Quality Mapping of Chandigarh Tricity Area:**
 - Designed and deployed low cost **IoT Air Quality Monitors** to create a dense Air Quality Map of the Chandigarh Tricity area.
 - Developed a custom **C++ sensor library** supporting 12 sensors, and 3 communication protocols with dynamic runtime sensor initialization.
 - Built an interactive web dashboard to visualize local air quality analytics for deployment on the **Chandigarh Citizens Foundation** website

Summer Reserach Intern, Prof. Sunita Chauhan, Plaksha University Jun 2025 - Aug 2025

- Developed the **UGV subsystem** of a collaborative UAV-UGV precision agriculture platform for autonomous crop monitoring.
- Built a **ROS 2/Gazebo digital twin** of a UGV and 4-DoF manipulator payload and tuned appropriate ROS Controllers.
- Deployed a **full navigation stack** - SLAM, localization, path planning - with **LiDAR and UAV imagery maps** on a testbed UGV.

Publications

Kapoor A, **Dayma S**, Narula M, Srinivasan S, *A Framework for High-Resolution Air Quality Monitoring for the Chandigarh Tricity Region*, 15th International Conference on Air Quality – Science and Application (AQ2026), Prague, June 2026. [doi:10.5281/zenodo.23179131](https://doi.org/10.5281/zenodo.23179131)

Projects

Experimental Modelling of Fluid Flow in Arteries | Prof. Shashikant Pawar Aug 2026 - Present

- Designed experimental setup to study fluid pressure and velocity and tube deformations in **arterial blood flow**.
- Utilising **physics informed ML for inverse modelling** and OpenFOAM CFD pipeline for extrapolating experimental measurements.

Comparison of Neural Network and RL based Controls | Prof. Sandeep Manjanna Feb 2026 - May 2026

- Comparison of **physics informed neural network (PINN)** based HJB-PDE solvers and RL algorithms for the inverted pendulum control problem.
- Constructed inverted pendulum system for hardware testing of policies, achieving **<10 degrees of tracking error** with naive PD position control.
- Determined PINN based controllers yield more optimal solutions with **<0.002 mean reward deviation from traj opt.** benchmark.

FOC Motor Control | Prof. Srikant Srinivasan Apr 2026 - May 2026

- Implemented **field oriented control** for a gimbal BLDC motor on **STM32** enabling **torque control** mode.
- Generated center aligned **sinusoidal PWM** via **Clarke and Park transforms**, with quadrature encoder feedback and startup rotor alignment.
- Utilised **inline current sensing** with timer-triggered ADC sampling, enabling feedback based vector motor control.

Digital Guitar Pedal | Prof. Srikant Srinivasan Feb 2026 - May 2026

- Engineered a custom **16-bit, 44.1kHz digital effects pedal** for the guitar from scratch, alongside an enclosure and full UI Interface.
- Designed **single supply biasing op-amp circuits** for pickup signal level and impedance matching with downstream ADC.
- Implemented **real-time, low-latency audio processing** on the Teensy 4.1, including a 10-stage tone stack, distortion, delay, reverb, etc.

Optimal Control using PINNs | Suraj Kumar, URSC, ISRO Sep 2025 - Dec 2025

- Developed physics informed neural networks to solve infinite horizon **optimal control problems** such as **inverted pendulum** and **cart pole**.
- Leveraged theory of functional connections and extreme learning machine to **solve Hamilton-Jacobi-Bellman PDE**.
- **Proposed XTFC-Unfreeze** model with **LQR initialization** showcasing improved non-linear control and stability for standard systems.

Lighter Than Air Systems | Prof. Rajkumar Pant Jan 2025 - May 2025

- Designed and fabricated a number of lighter than air systems such as **aerostat, airship envelopes, recovery devices, and launchers**.
- Engineered and deployed a **glider launch mechanism on an aerostat**, achieving a **30m+ glide from a 10m altitude**.
- Designed and validated an **emergency rapid deflation system** for airships, enabling both payload and envelope material recovery.

Skills

Programming Languages: Python, C, C++, SQL, Bash, Assembly (ARM Cortex-M, MIPS)
Environments: ROS2, Gazebo Simulator, MuJoCo, MJX
Math & Optimisation: Linear Algebra, Dynamics, Trajectory Optimisation, Model Predictive Control
Other Skills: Git, $\LaTeX$, CAD (Fusion 360, Onshape), CFD (COMSOL, SimScale, OpenFOAM)

Relevant Coursework

RO2001: System Dynamics and Controls Prof. Sunita Chauhan	Grade: A+
RO3003: Control Autonomy Planning and Navigation Prof. Shashank Tamaskar	Grade: A+
RO3101: Robot Motion, Simulation, and Optimisation Prof. Andy Ruina	Grade: A
CP3001: Embedded Systems Prof. Srikanth Srinivasan	Grade: A
CP2002: Signals and Systems Prof. Vivek Deulkar	Grade: A
ME3001: Engineering Mechanics Prof. Shashikant Pawar	Grade: A
ME3002: Fluid Mechanics Prof. Shashikant Pawar	Grade: A

Teaching

Student Tutor, FM113: Foundations of Physical World Plaksha University	Aug 2026 - Present
- Supporting instruction for freshman physics course with 300+ students, under Prof. Andy Ruina, Prof. Shashikant Pawar and Prof. Praveen Kumar. - Conducting tutorial sessions related to statics, dynamics, work and energy, and electromagnetism.	
Student Tutor, FM214: Intelligent Machines Plaksha University	Aug 2026 - Present
- Supporting instruction for sophomore robotics course with 230+ students, under Prof. Andy Ruina and Prof. Sandeep Manjanna. - Conducting tutorial sessions related to circuit analysis, DC motors, sensors, control and estimation, and planning.	
Student Tutor, FM206: Mechatronics Plaksha University	Aug 2025 - Dec 2025
- Supported instruction for a sophomore mechatronics course with 180+ students, under Prof. Andy Ruina. Designed demonstrations to teach concepts like ADCs, circuit analysis, DC motors, servo motors, PWM, odometry and, forward kinematics. - Assisted in conducting tutorials and laboratory sessions, ensuring safe practices, debugging hardware, and helping students build intuition.	

Positions of Responsibility

Cohort Representative, Robotics and Autonomous Systems Plaksha University	Jul 2026 - Present
- Represented RAS cohort, facilitating communication between 20+ faculty members and the student body. - Bridging student-faculty communication on course policies resolving student academic challenges through continuous feedback.	
Co-organiser, TEDxPlakshaUniversity	Jan 2024 - Apr 2024
- Launched the first TEDx event at Plaksha University, directing a 10+ member team across speaker curation, production and outreach. - Curated 5 speakers across disciplines; recorded talks have since drawn 100k+ views on YouTube.	

Achievements

- Abu Dhabi Student Golden Visa:** Long-term UAE residence visa awarded for academic excellence.
- Merit Scholarship, Plaksha University:** Recognized for academic excellence and extracurricular contributions.
- Academic Excellence Award** from Rochester Institute of Technology, Dubai.